COURSE FINAL PROJECT · IMPLEMENTATION REPORT

Airfoil Explorer

DRL × LBM Design Space

PARTICIPANT
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SECOND PARTICIPANT
Pending — not supplied

DATE
10 August 2026

STATUS
Submission draft; production evidence pending

Problem and required achievement

The thesis workflow uses lattice Boltzmann analysis to evaluate airfoils and deep reinforcement learning to explore BP3333 geometry changes. The result set is scientifically useful but difficult to review interactively: geometry and aerodynamic coefficients must stay synchronized, while fields not approved for public release remain outside the deployed artifact.

Defined achievement

A public synthetic-fixture demonstrator in which a reviewer drags geometry parameters, sees the nearest stored wing and its C_l/C_d update live, and releases to snap every control to that exact row. The public contract is limited to its declared geometry, coefficient, validity, and provenance fields. The interface never predicts or interpolates aerodynamic values.

Why this is an engineering task

- **Scientific consistency:** the wing and every displayed coefficient must originate from one admitted provenance row.
- **Interaction quality:** nearest-neighbor search must remain deterministic, fast, accessible, and stable during drag.
- **Publication safety:** credentials, local paths, W&B namespace, and undeclared diagnostics remain outside the public artifact.
- **Operational safety:** ordinary deployment must be secret-free and fixture-only; optional live publication needs an explicit gate.

Safety model

The selected delivery target is a **public GitHub Pages** project site, not an authentication or email-allowlist boundary. Main/manual deployment replays only the committed fixture and needs no secret. A separate hourly/manual W&B workflow remains disabled unless `PUBLIC_RESEARCH_DATA_APPROVED=true` and an API key is present; enabling it makes the declared public contract available. URL: edengol10.github.io/AI-final-course/. References: GitHub Pages workflows, Pages visibility, and W&B Public API.

Repository boundary. The thesis checkout remained a read-only scientific reference at commit `4936dea`; production code neither imports nor mutates it. All implementation lives in the separate private `AI-final-course` repository.

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Product and architecture

A static, credential-free client backed by validated, immutable data snapshots.

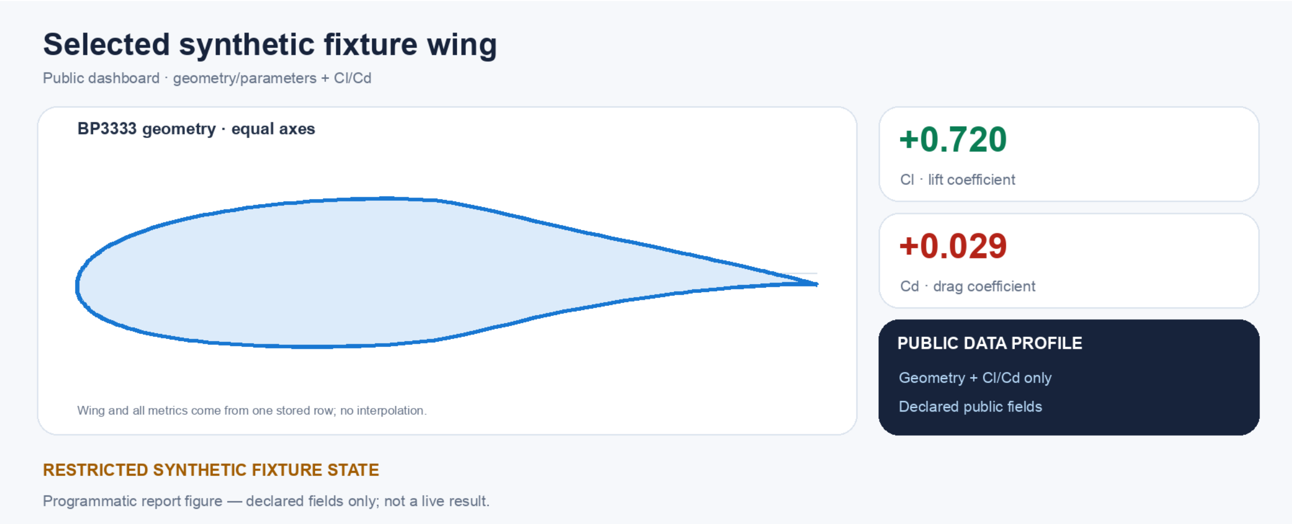

Figure 1. Representative UI state reconstructed from the repository's synthetic QA fixture.

7	8	5	6
unique fixture geometries	admitted fixture samples	reviewed run identities	audited fixture rejections

Data path and trust boundary

Committed fixture → canonical JSON/SHA-256 → Vite subpath build and privacy scan → Pages artifact → public GitHub Pages → browser Zod/hash validation and Web Worker search. The optional approved W&B path performs narrow export and admission before joining the same publication path.

Interaction contract

- During pointer or keyboard drag, an animation-frame-throttled worker normalizes only variable parameters by authoritative BP3333 bounds and selects the exact nearest row.
- Ghost markers preserve requested positions; the wing, signed C_l/C_d bars, and provenance preview always change together.
- On release, all sliders animate to the selected database vector. Stable record index breaks equal-distance ties, preventing flicker.
- Refresh recomputes the manifest's canonical SHA-256, then each shard's byte size and SHA-256, and validates every schema before an atomic in-memory swap. Unchanged, malformed, stale, offline, and empty states remain distinct.

Public data contract

The manifest's explicit `snapshotKind` is either `synthetic-fixture` or `reviewed-wandb`; the interface uses it to label fixture evidence visibly. The approved scientific outputs are BP3333 geometry/parameters and per-wing C_l/C_d. Minimal run/step provenance, replicate count, curvature/admission metadata, compatibility descriptors, bounds, counts, and checksums remain for auditability; W&B entity/project names and per-row coordinate arrays are absent.

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AI co-coding process and skill evidence

AI accelerated implementation, while domain contracts and human review controlled scientific risk.

Same-prompt before/after experiment

> **"Build the W&B-to-dashboard data refresh and nearest-wing interaction while preserving scientific validity and protecting unpublished data."**

A prospective unskilled design was frozen before the custom skill was read. The identical prompt was then answered with the *curate-aero-wandb-data* skill. Each output was scored 0–5 against six predeclared criteria; both artifacts and the rubric are preserved in `experiments/skill-comparison/`.

Criterion	Without	With	Observed correction
Secret exposure	4/5	5/5	Exact narrow-history and exclusion boundary
Per-wing keys	2/5	5/5	Pinned step C_l/C_d values
Invalid rejection	3/5	5/5	Ordered, stable reason codes
Physical grouping	3/5	5/5	Exact dimensions; unknowns isolated by run
Provenance	4/5	5/5	Minimal run/step/time tied to one metric tuple
Reproducibility	3/5	5/5	Authoritative bounds and deterministic representative
Total	**19/30**	**30/30**	**Domain rules became testable contracts**

Material human corrections

- Rejected observed-range normalization: the nearest wing would change when sample coverage changed. Fixed authoritative BP3333 bounds now define distance.
- Reduced the published schema to the explicitly approved geometry and coefficient fields.
- Removed an early per-row coordinate-array field; TypeScript reconstructs 253 geometry points from the ten parameters.
- Replaced an analytic reference curve with the exact BP3333 NACA2412 baseline; split unique-wing and successful-sample counts.

Agent split and evidence discipline

The lead established the immutable thesis boundary and integration contract. Independent data, frontend, and QA agents worked in parallel, followed by lead review. The portable project skills `curate-aero-wandb-data` and `verify-airfoil-explorer` encode future refresh and acceptance workflows. Including hosting conversion, the evidence log's conservative line items total about 24 hours 50 minutes; this report rounds down to **24 hours saved**, with scientific judgment and release approval remaining human responsibilities.

> **Observed value of the custom skill.** The major improvement was not writing speed; it was eliminating plausible scientific errors in field choice, normalization, grouping, and replicate provenance before data crossed the publication boundary.

Validation, limitations, and reflection

Passed evidence is separated from execution-blocked work; fixture results are never presented as live research results.

Area	Evidence	Status
Exporter	40 Python tests; deterministic fixture, grouping, rejection, and declared profile	Passed
Frontend	16 tests; ESLint; strict TypeScript; production Vite build	Passed
Geometry	253 finite points; Python/TypeScript max error $\approx 1.2 \times 10^{-7} < 10^{-5}$	Passed
Privacy	Scanner self-test and built `dist/` scan; no secret/path match	Passed
Project skills	Both folders pass the official structural validator	Passed
Playwright	Nine cases discovered/type-checked; loopback server denied	Blocked
Live W&B	Client communication failed closed; fixture was not replaced	Blocked
GitHub Pages	Workflow deployed; public URL verified by HTTP smoke check	Passed

Physical validity and limits

Admission requires a finished approved run, known absolute-action schema, bounded finite actions, finite raw step C_l/C_d, valid geometry, and curvature ratio below one. Incompatible baseline, AoA, averaging window, or solver revision values never mix; unknown settings are isolated. Exact float32 vectors define unique geometries, while every valid replicate remains counted and traceable.

The committed source snapshot is deliberately synthetic: `snapshotKind=synthetic-fixture`, 8 admitted samples, and 7 geometries across five reviewed run identities. The Pages workflow rebuilds it before publication. It proves the software contract, not thesis performance. The public URL is deployed; no claim is made that live W&B export works.

Reflection

Achieved: fixture mode demonstrates the defined interaction, one-row coefficient consistency, explicit fixture provenance, checksum-verified loading, and fail-closed refresh behavior. **Pending:** live W&B export remains optional and unproven. AI produced substantial breadth quickly, while review kept scientific and publication decisions human-controlled. **In hindsight,** I would define the public field profile, `snapshotKind`, provenance redaction, subpath build, branch/environment protection, and deployment evidence gate before the first implementation batch.

Submission handoff

Before final submission

- Insert the exact second participant name.
- Enable Settings → Pages → GitHub Actions.
- Protect `main` and require fixture CI before merge.
- Deploy and verify the expected URL plus public JSON paths.
- Confirm the artifact and UI contain only declared public fields.
- Run browser/mobile smoke tests and attach screenshots/video.
- Replace the source QR only after the Pages URL is verified.

Private source repository

github.com/edengol10
AI-final-course